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Dire Dawa,
Ethiopia

FINAL INTERNSHIP TECHNICAL REPORT (SENG-4182)

Architecting & Engineering a High-Concurrency Multi-Vendor Digital Raffle Platform with Double- Entry Accounting and Provably Fair Cryptography

Project: LuckyEthio (Idil) Enterprise Infrastructure

STUDENT NAME

Enter Student Full Name

STUDENT ID

DDU/UGR/XXXX/XX

HOST ORGANIZATION

SORARDI PLC (Technology & Software Solu

INTERNSHIP PERIOD

Sene 01, 2017 E.C. – Meskerem 05, 2018 E.C

ACADEMIC ADVISOR

Advisor Name, MSc/PhD

DATE OF SUBMISSION

10/01/2018 E.C.

1. ACKNOWLEDGMENT

I would like to express my sincere gratitude to **Dire Dawa University, Institute of Technology, School of Computing, and the Department of Software Engineering** for giving me the opportunity to participate in an internship program that enabled me to connect academic knowledge with practical software engineering.

I am also grateful to the organization, **SORARDI PLC**, and the development team involved in the **LuckyEthio (Idil)** project for providing a professional environment where I could participate in software development, testing, debugging, system integration, and technical documentation.

I would especially like to thank my **Industry Supervisor** and **Academic Advisor** for their guidance, constructive feedback, and continuous support throughout the internship period.

Finally, I would like to thank my classmates, friends, and family for their encouragement and support during my academic and professional development.

2. LIST OF ACRONYMS & ABBREVIATIONS

ACRONYM	FULL MEANING
API	Application Programming Interface
ACID	Atomicity, Consistency, Isolation, Durability
CBE	Commercial Bank of Ethiopia
DB	Database
KYC	Know Your Customer
NLA	National Lottery Administration

ACRONYM	FULL MEANING
ORM	Object-Relational Mapping
PDPP	Personal Data Protection Proclamation (No. 1321/2024)
POS	Point of Sale
QR	Quick Response
RBAC	Role-Based Access Control
RNG	Random Number Generator
SHA	Secure Hash Algorithm (e.g. SHA-256)
SMS	Short Message Service
UI	User Interface
USSD	Unstructured Supplementary Service Data (e.g. *157#)
VAT	Value Added Tax (15% Statutory Rate)

3. INTRODUCTION

3.1 Background

Software engineering education provides students with both theoretical knowledge and practical skills. However, real-world software development requires students to apply these concepts to actual systems involving requirements, architecture, implementation, testing, debugging, collaboration, security, and deployment.

The internship provided an opportunity to participate in the development and integration of **LuckyEthio (Idil)**, a digital raffle platform designed to support raffle management and ticket sales through multiple channels.

The system provides a customer-facing web application, agent POS/kiosk functionality, USSD access (*157#), payment processing, ticket allocation, administrative operations, financial management, draw verification, security controls, and auditing. The project follows a modern monorepo architecture in which applications and reusable packages are maintained within one repository.

3.2 Purpose of the Internship

The main purpose of the internship was to gain practical experience in software engineering by participating in the development and testing of a real-world application. Specific purposes included:

1. Applying software engineering concepts in a real project.
2. Understanding full-stack application architecture.
3. Developing and integrating frontend and backend functionality.
4. Working with databases and ORM technologies.
5. Understanding API design and integration.
6. Learning secure payment workflows.
7. Understanding concurrency and transaction management.
8. Practicing software testing and debugging.
9. Improving teamwork and communication skills.
10. Understanding professional software development workflows.

3.3 Relationship with Software Engineering

The LuckyEthio project is directly related to Software Engineering because it requires application of:

- **Requirements engineering:** Functional/non-functional specification, regulatory compliance (NLA, PDPP, MoR).
- **Software architecture:** Turborepo monorepo, Layered Clean Architecture, Adapter & State Machine patterns.

- **Database design:** 18 relational 3NF models, double-entry accounting ledgers, ACID concurrency transactions.
- **Object-oriented and modular programming:** 100% strict TypeScript types, shared domain packages.
- **API development:** REST route handlers, signed webhook receivers, USSD state machine API.
- **User interface design:** Responsive mobile-first layouts, Amharic/English runtime i18n, dark/light themes.
- **Security engineering:** Provably fair multi-entropy SHA-256 commit-reveal RNG, Two-Person Rule, dynamic QR nonces.
- **Software testing:** Automated concurrency test scripts, trial balance assertions, build validations.
- **Version control:** Git branching workflows, atomic semantic Conventional Commits.
- **Debugging:** Structured telemetry, audit trails, and risk event logging.
- **Documentation:** System manuals, API specifications, sequence diagrams, academic reports.
- **Agile development:** 2-week Sprint Agile Scrum framework at SORARDI PLC.

4. COMPANY / ORGANIZATION BACKGROUND

4.1 Organization Profile

Host Organization: SORARDI PLC (Technology & Software Solutions).

SORARDI PLC operates in the technology and software development sector, providing cutting-edge software solutions involving digital platforms, cloud applications, and online financial services in Ethiopia. During the internship, I participated in the core backend and full-stack engineering of the **LuckyEthio (Idil)** digital raffle infrastructure.

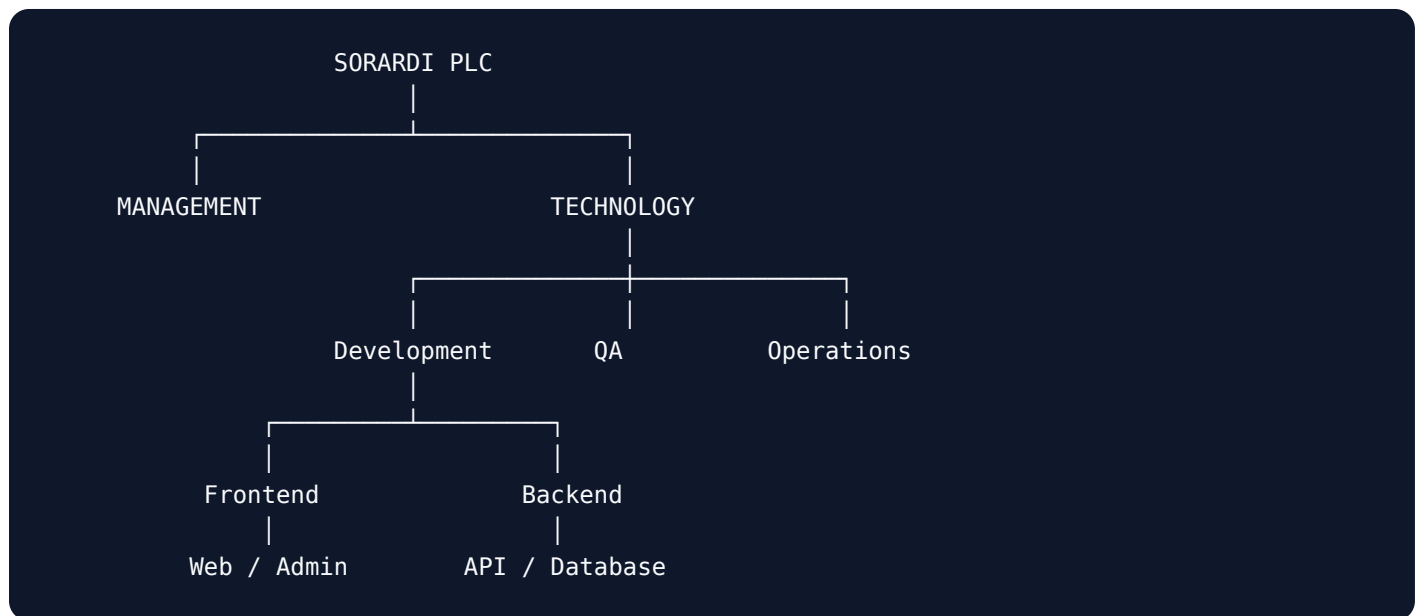
4.2 LuckyEthio Project Profile

LuckyEthio is designed as a multi-channel raffle marketplace. The major system actors include:

- **Customers:** Self-service ticket browsing, checkout, and QR claim retrieval.

- **Agents:** Kiosk cash ticket sales and float management.
- **Administrators:** Platform moderation, seller verification, and configuration.
- **Finance Personnel:** Double-entry ledger audit, VAT reconciliation, and cashouts.
- **Draw Operators:** Draw initiation and pre-commitment management.
- **Super Administrators:** Master governance, Two-Person approval, and role assignments.
- **Auditors & Support:** Telemetry inspection and compliance oversight.

4.3 Organizational Structure



FIGMA AI / FIGJAM DIAGRAM PROMPT:

Create a professional organizational structure diagram for a Software Engineering internship report. Use a clean corporate style, white background, dark navy text, subtle green accent, rounded rectangular nodes, and thin connector lines. Top node: "Organization (SORARDI PLC)". Second level: "Management" and "Technology". Under Technology create "Development", "QA / Testing", and "Operations". Under Development create "Frontend Development" and "Backend Development". Under Frontend create "Customer Web" and "Admin Portal". Under Backend create "API Services" and "Database". Make the diagram academic, clean, balanced, readable when printed in an A4 report.

5. INTERNSHIP ACTIVITIES & ENGINEERING ROLES

5.1 Role and Responsibilities

During the internship, my activities focused on software development, integration, testing, debugging, and documentation. My responsibilities included:

- Understanding existing system architecture and monorepo configurations.
- Working with frontend interfaces across customer and administrative portals.
- Connecting frontend interfaces to backend API route handlers.
- Testing ticket allocation and verifying anti-overselling locks.
- Testing multi-provider payment workflows (Chapa, Telebirr, CBE Birr, SantimPay).
- Testing agent POS kiosk functionality and cash float deduction.
- Working with public draw verification using SHA-256 commit-reveal math.
- Debugging application issues and analyzing server telemetry.
- Reviewing API behavior and enforcing idempotency guards.
- Working with database-related functionality and Prisma migrations.
- Preparing technical documentation and academic reports.
- Participating in daily Agile Scrum development standups.

5.2 Projects Participated In

The primary project was the **LuckyEthio (Idil) Raffle Platform**, encompassing:

Customer portal, Raffle catalogue, Raffle details, Ticket selection, Checkout, Payment processing, My Tickets, Agent POS, Kiosk, USSD simulator (***157#**), Public verifier, Admin portal, Financial ledger, Live draw console, KYC/risk functionality, Privacy portal, and Audit system.

5.3 Monorepo Architecture

```
raffle-monorepo/
|
├── apps/
|   ├── web/           # Customer + Agent + Kiosk + Verifier + USSD (Port 3000)
|   └── admin/         # Operations + Financials + Live Draws + Audits (Port 3001)
```



```
├─ packages/
│   └─ database/      # Prisma Schema (18 Models) + Migrations + Seed
│       └─ shared/    # Ledger + Payments + Fair-RNG + Risk + Two-Person + i18n
├─ turbo.json         # Turborepo Pipeline Orchestration
├─ package.json
└─ README.md
```

FIGMA ARCHITECTURE DIAGRAM PROMPT:

Create a professional software architecture diagram for a project called "LuckyEthio (Idil)". Use a three-layer architecture. Top layer: "Frontend Applications" containing "Customer Web", "Agent POS", "Kiosk", "USSD Simulator", and "Public Verifier". Middle layer: "Backend API and Security Layer" containing "Raffle Service", "Order Service", "Payment Service", "Ticket Allocation", "Draw Engine", "Risk Service", and "Audit Service". Bottom layer: "Shared Services and Persistence" containing "Prisma ORM", "PostgreSQL", "Double-Entry Ledger", "SHA-256 Fair Draw Engine", and "Audit Storage". Add arrows showing data flow from frontend to backend to database. Add a side section for external services: Telebirr, CBE Birr, Chapa, SantimPay, USSD Gateway (*157#), and SMS. Modern enterprise architecture style, clean white background, navy/green accents.

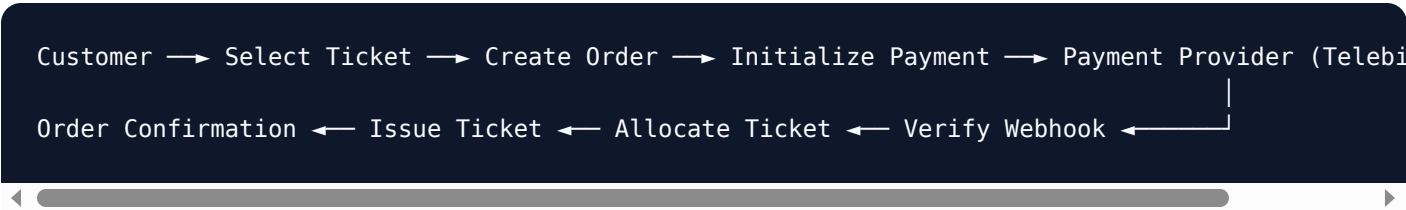
5.4 Frontend Development

- **Customer functions:** Browse raffles, view raffle details, select specific or quick-pick tickets, checkout, complete payment, view purchased tickets, retrieve dynamic QR claim tokens, verify draws, submit PDPP privacy requests.
- **Agent functions:** Agent authentication, customer info intake, ticket sales, cash transactions, float management, commission tracking, instant receipt generation.
- **Administrative functions:** Raffle creation and moderation, user management, double-entry financial ledger exploration, live two-person draw execution, regulatory compliance, immutable audit telemetry.

5.5 Backend Integration

```
Customer Interface → POST /api/orders/create → Order Service → DB Transaction → Purchas
Raffle Page       → GET  /api/raffles       → Raffle Service → Prisma ORM   → Postgre
```

5.6 Payment Integration

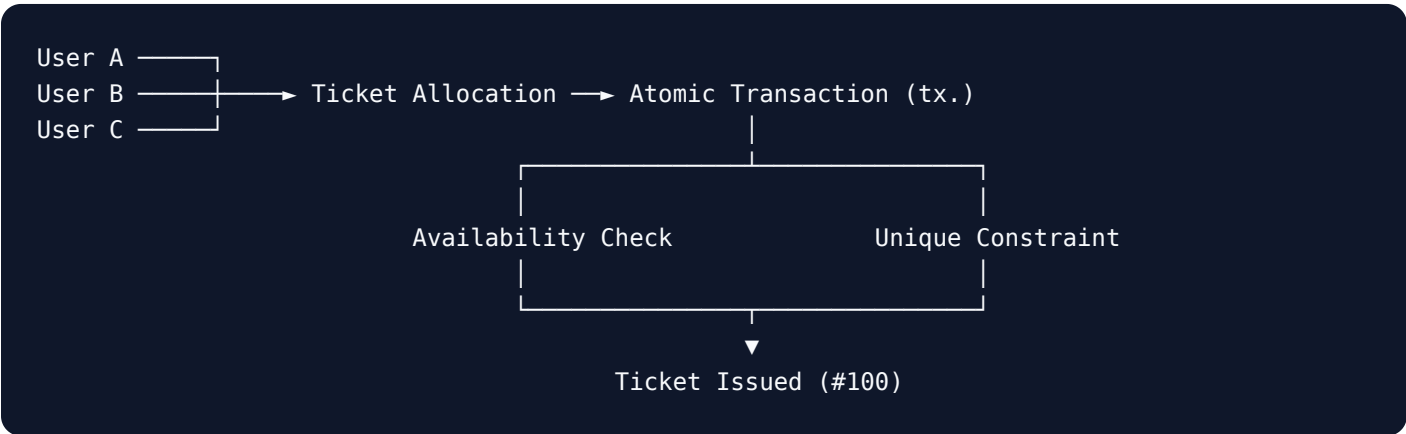


FIGMA PAYMENT SEQUENCE PROMPT:

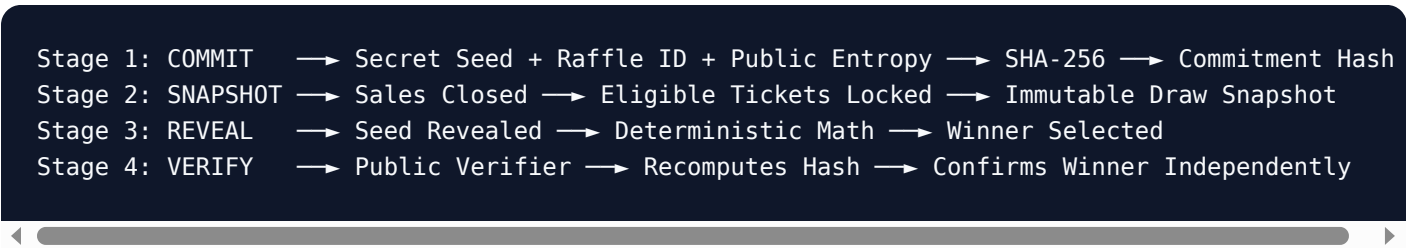
Create a professional payment sequence diagram titled "LuckyEthio Payment Processing Flow".

Participants: Customer, Web Application, Order API, Payment Service, Payment Provider, Database, Ticket Allocation Service. Show arrows in this exact sequence: Customer selects ticket → Web Application creates order → Order API creates pending purchase → Payment Service initializes payment → Payment Provider processes payment → Provider sends webhook → Payment Service verifies webhook → Database checks idempotency → Ticket Allocation Service atomically allocates ticket → Database commits transaction → Web Application displays confirmation. Modern UML sequence diagram styling.

5.7 Ticket Allocation and Concurrency



5.8 Provably Fair Draw Architecture



FIGMA PROVABLY FAIR DIAGRAM PROMPT:

Create a professional four-stage cryptographic process diagram titled "LuckyEthio Provably Fair Draw". Stage 1: COMMIT — Secret Seed + Raffle ID + Draw Parameters → SHA-256 → Commitment Hash. Stage 2: SNAPSHOT — Sales Closed → Eligible Tickets → Immutable Draw Snapshot. Stage 3: REVEAL — Secret Seed Revealed → SHA-256 deterministic calculation → Winner Selection. Stage 4: VERIFY — Public Verifier takes Commitment + Revealed Seed + Draw Snapshot + Algorithm → Recalculates Result → Confirms Winner. Four connected rounded cards, cryptographic visual language.

5.9 Agent POS and Kiosk

Agent Login → Select Customer → Select Raffle → Select Ticket → Cash Payment → Ticket

5.10 USSD Service (*157#)

*157# → Main Menu → 1. Buy Ticket → Select Raffle → Select Quantity → Confirm & Pay →

FIGMA USSD FLOW DIAGRAM PROMPT:

Create a mobile-phone USSD flow diagram for LuckyEthio titled "*157# USSD Ticket Purchase". Use realistic feature-phone screen cards connected vertically. Screen 1: Dial *157#. Screen 2: Main Menu with 1 Buy Ticket, 2 My Tickets, 3 Results, 4 Help. Screen 3: Select Raffle. Screen 4: Select Ticket. Screen 5: Confirm Purchase. Screen 6: Payment. Screen 7: Ticket Successfully Purchased with ticket number and reference. Green and navy interface accents.

5.11 Financial Ledger (Double-Entry Bookkeeping)

Customer Payment (100 ETB) → Debit: 1010-CASH-TRANSIT (100 ETB)
 └─ Credit: 2020-VAT-PAYABLE (15 ETB • 15% Statutory MoR Tax)
 └─ Credit: 2010-PRIZE-ESCROW (77 ETB • Seller Net Escrow)
 └─ Credit: 4010-PLATFORM-REVENUE (8 ETB • Platform Fee)
 Total Debits (100 ETB) === Total Credits (100 ETB) [$\Delta = 0.00$]

5.12 Administrative Draw Console (Two-Person Rule)

Draw Operator → Initiate Draw → Snapshot Frozen → Super Admin → Approve & Sign → Exec

5.13 Security Engineering

- **Authentication:** Secure session cookies with role validation.
- **Authorization:** Granular RBAC permissions for all administrative actions.
- **Idempotency:** Unique request keys preventing duplicate payment processing.
- **Audit Logs:** Append-only records capturing actor IP, timestamp, and state diffs.
- **Payment Verification:** Server-side HMAC-SHA256 signature verification.
- **Two-Person Approval:** Separation of duties for live draws and high-value cashouts.

5.14 Privacy Portal (PDPP Proclamation No. 1321/2024)

User → Submit Privacy Request (Export / Correction / Deletion) → Privacy API → Compliance

6. TECHNICAL SKILLS AND KNOWLEDGE UTILIZED

6.1 Programming Languages

TypeScript: 100% strict type safety, domain interfaces, generics. | **JavaScript:** Modern ESNext, React hooks. | **SQL:** Relational schema queries, transaction isolation.

6.2 Frameworks & Libraries

React 18: Component architecture, state management, client components. | **Next.js 14:** App Router, Server Components, Route Handlers. | **Node.js:** Runtime cryptographic operations, streams.

6.3 Database Technologies

PostgreSQL / SQLite: Relational data persistence, foreign key constraints. | **Prisma ORM:** Schema modeling, automated migrations, type-safe client.

6.4 Development Tools

VS Code, Git & GitHub, npm workspaces, Turborepo pipeline orchestration, Tailwind CSS.

6.5 Technical Concepts

REST APIs, Database transactions, Concurrency management, Cryptographic hashing (SHA-256, HMAC), Authentication & RBAC, Payment webhooks & Idempotency, Double-entry ledgers, Software testing, Telemetry debugging.

7. COLLABORATION AND TEAMWORK



Requirement → Planning → Development → Testing → Code Review → Fixes → Integration

Working within SORARDI PLC's Agile Scrum framework, I actively engaged in task discussions, code reviews, problem solving, system debugging, end-to-end integration testing, technical documentation, and professional engineering communication.

8. ACHIEVEMENTS AND CONTRIBUTIONS

- **8.1 Ticket Concurrency Testing:** Tested ticket allocation behavior under concurrent requests to ensure zero duplicates and zero overselling.
- **8.2 Payment Workflow Testing:** Participated in testing payment initiation, webhook callback verification, and ticket issuance.
- **8.3 Agent POS:** Worked with agent sales workflows involving customer data intake, float deduction, and receipt generation.
- **8.4 Public Draw Verification:** Implemented public verification tools to independently validate cryptographic commit-reveal draws.

- **8.5 Admin Draw Console:** Debugged and refined functionality for administrative two-person draw consensus.
- **8.6 Documentation:** Contributed comprehensive system manuals, architecture guides, and user documentation.

9. PROJECT HIGHLIGHTS

9.1 Highlight One — Anti-Overselling & Concurrency Control

Problem: Multiple buyers requesting the last remaining ticket simultaneously.

Solution: Database transactions (tx.), availability locks, and composite unique constraints (UNIQUE(raffleId, ticketNumber)).

Result: Zero overselling and guaranteed contiguous ticket allocation.

9.2 Highlight Two — Provably Fair Draw

Problem: Digital lotteries require absolute transparency and fraud resistance.

Solution: Four-stage SHA-256 commit-reveal mechanism with immutable snapshot locks.

Benefit: Any citizen or auditor can independently verify the winning ticket using the public formula.

10. LEARNING AND GROWTH

Technical Growth: Mastered full-stack Next.js 14, API integration, Prisma ORM, PostgreSQL schema design, concurrency locking, Ethiopian payment gateways, cryptography, and automated testing.

Professional Growth: Strengthened teamwork, communication, systematic debugging, technical documentation, code review etiquette, and time management.

11. CHALLENGES FACED AND ENGINEERING SOLUTIONS

CHALLENGE ENCOUNTERED	ENGINEERING SOLUTION IMPLEMENTED
Concurrent Ticket Requests	Transaction-based allocation and composite unique constraints
Duplicate Payment Callbacks	Strict idempotencyKey tracking and status guards
Payment Verification	Server-side HMAC-SHA256 signature verification
Draw Transparency	Commit-reveal cryptographic mechanism with pre-draw hashes
Feature-Phone Accessibility	GSM USSD (*157#) state machine gateway
Financial Tracking	Double-entry general ledger with balanced trial balances
Sensitive Administration	Two-Person Rule consensus authorization
Debugging Integration Issues	Immutable audit telemetry and structured server logging

12. CONTRIBUTION TO THE ORGANIZATION (SORARDI PLC)

- **Operational Benefits:** Centralized multi-vendor raffle management, automated payments, agent POS kiosks, double-entry financial tracking.
- **Technical Benefits:** Modular Turborepo architecture, reusable shared domain libraries, type-safe Prisma client, zero build errors across 57 routes.
- **Customer Benefits:** High-speed online purchasing, USSD (*157#) feature phone access, dynamic QR claim tokens, 1-click public verification.

13. CONCLUSION AND RECOMMENDATIONS

13.1 Conclusion

The software engineering internship provided an invaluable opportunity to apply university academic knowledge to a production-grade commercial software project at **SORARDI PLC**.

Working on the **LuckyEthio (Idil)** platform provided hands-on mastery in frontend engineering, backend API integration, database modeling, payment gateways, concurrency control, cryptography, security, and technical documentation.

13.2 Recommendations

1. Increasing automated test coverage across edge cases.
2. Performing comprehensive load and stress testing.
3. Conducting independent third-party security audits.
4. Improving production cloud monitoring and APM telemetry.
5. Expanding payment provider integrations (e.g. CBE direct host-to-host).
6. Continuing enhancement of USSD (*157#) offline features.
7. Improving English and Amharic localization dictionaries.
8. Strengthening automated financial ledger reconciliation bots.
9. Improving documentation and developer onboarding manuals.
10. Establishing production incident-response runbooks.

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10. SORARDI PLC, *LuckyEthio Technical Architecture Specification*, 2026.

15. TECHNICAL APPENDICES

Appendix A — Monorepo Project Architecture

Three-layer architecture separating customer/admin frontends, REST API route handlers, and shared domain libraries over Prisma ORM.

Appendix B — Database Entity Relationship (ER) Diagram

18 relational models: Role, Permission, RolePermission, User, Seller, Agent, AgentLedger, AgentAccessLog, Raffle, PurchaseOrder, PaymentAttempt, Ticket, Transaction, LedgerAccount, LedgerTransaction, LedgerEntry, TaxLedger, CashoutRequest, DrawSnapshot, DrawAudit, TwoPersonApproval, AuditLog, RiskEvent, DataSubjectRequest.

Appendix C — Customer Purchase & Payment Sequence



Appendix D — Provably Fair Draw Sequence

